◆ TEAM TRANSIENCE

FUGACITY 2026 · IIT KHARAGPUR · PHASE 1 SUBMISSION

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क्षणिक Kshanik

"kṣaṇik" — transient, momentary · Physics-Informed Yield Surrogate for a Non-Isothermal Plug-Flow Reactor

> A machine learning surrogate for reactor yield prediction, built by fitting the governing reaction kinetics and energy balance directly, then applying **Gaussian Process** and **CatBoost** models to correct only the residual error the mechanistic fit leaves behind.
>
> **Cross-validated RMSE 8.79 on 150 training observations**

Team Transience • Fugacity 2026 ML Hackathon, IIT Kharagpur

• Vaishnav Aditya · Preenithi Varshana · Kritika Pandey • 150 training rows, 50 held-out test rows


```
train_dataset.csv → mechanistic_pfr.py → warped GPR / CatBoost → overall_yield
```

GitHub Repository ↗ | Full Technical Report (PDF) ↗

25%	5 → 13	8.79	2
ROWS AT EXACT ZERO YIELD	RAW INPUTS → PHYSICS FEATURES	BEST CV RMSE (WARPED GPR)	MODELS: SMALL-DATA & LARGE-DATA

THE CORE PROBLEM VS. THE APPROACH

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LIMITATION OF DIRECT REGRESSION

Black-Box Models Cannot Isolate the Yield Collapse

A regressor fit directly on `flow_rate`, `concentration`, `inlet_T`, `length`, and `jacket_T` must infer, from 150 rows alone, that yield falls to exactly **0%** whenever residence time and temperature drive the side reaction (B→C) past the desired one (A→B). No single raw feature correlates strongly with yield — the highest is **|r| = 0.64** — because the governing variable is a non-linear combination the model has no basis to construct from data this limited.

TRANSIENCE APPROACH

A Mechanistic Prior, Corrected by a Gaussian Process

We fit a non-isothermal PFR model directly — series kinetics **A→B→C** under Arrhenius rate laws, coupled to a jacket energy balance — using nonlinear least squares on the training set. Its output serves as the **prior mean** for a Gaussian Process, which is left to learn only the mechanistic model's residual error rather than the full yield surface.

MODEL PERFORMANCE — 5×10 REPEATED K-FOLD CV

MECHANISTIC MODEL ALONE	ADDITIVE-RESIDUAL GPR	WARPED-LOGIT GPR	CATBOOST + SYNTHETIC
12.17	10.14	8.79	11.35
RMSE · physics only, no ML	RMSE · mechanistic mean + plain GP	RMSE · selected small-data model	RMSE · large-data model

HOW IT WORKS — 5-STAGE MODELING PIPELINE

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1. **FIT ODE**
   Arrhenius kinetics and energy balance, fit by least squares on the training set

2. **ENGINEER FEATURES**
   Damköhler numbers, selectivity ratio, residence time, ΔT derived from the fit

3. **WARP TARGET**
   Logit transform of yield/100, respecting the [0, 100] boundary

4. **LEARN RESIDUAL**
   Matérn-kernel Gaussian Process fits the mechanistic model's error

5. **PREDICT**
   Inverse transform back to overall_yield for the test set

THE GOVERNING SYSTEM — FIT DIRECTLY TO TRAINING DATA

MOLE BALANCES

$$dC_A/d\tau = -k_1(T)\, C_A$$
$$dC_B/d\tau = k_1(T)\, C_A - k_2(T)\, C_B$$

ARRHENIUS KINETICS

$$k_i(T) = k_{0,i} \exp(-E_{a,i} / RT)$$

$i \in \{1, 2\}$ — one per reaction

ENERGY BALANCE

$$dT/d\tau = [U_a(T_j - T) + (-\Delta H_1) r_1 + (-\Delta H_2) r_2] / \rho C_p$$

jacket heat exchange + reaction enthalpy of both steps

Eight effective parameters — $k_{0,1}$, $E_{a,1}$, $k_{0,2}$, $E_{a,2}$, U_a, ΔH_1, ΔH_2, ρC_p — are recovered from the 150 training rows via multi-start nonlinear least squares, not assumed from literature values. The fitted result: $\mathbf{E_{a,2} > E_{a,1}}$, meaning the side reaction accelerates with temperature faster than the desired one — the direct physical cause of the zero-yield cliff seen in the training data.

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MECHANISTIC FOUNDATION

Non-Isothermal PFR Model

Series reaction A→B→C under Arrhenius kinetics $k_1(T)$, $k_2(T)$, coupled to a jacket energy balance with explicit reaction-heat terms (Fogler, *Elements of Chemical Reaction Engineering*, Ch. 8). Fit as a grey-box model via multi-start nonlinear least squares, not assumed from literature constants.

`Train RMSE:` **`12.17`** `· Ea₂ > Ea₁, consistent with side-reaction dominance at high temperature`

SMALL-DATA MODEL

Warped Gaussian Process

The GP is trained in logit(yield/100) space rather than on the raw residual. This is the statistically appropriate likelihood for a target bounded on [0, 100] with substantial mass at the boundary, consistent with the censored-regression and warped-GP literature.

`CV RMSE:` **`8.79`** `± 3.87 · outperforms both additive-residual and classifier-gated variants`

LARGE-DATA MODEL

CatBoost with Synthetic Augmentation

5,000 synthetic observations, sampled across the operating envelope and simulated through the fitted ODE, are combined with the real data at a 0.15× training weight. A classifier stage first separates the collapsed-yield regime before a regressor is applied, a strategy intended to scale to a larger production dataset.

`CV RMSE:` **`11.35`** `· versus 15.24 on real data alone, no augmentation`